COMP20008 Elements of Data Processing

Lecture 8: Distance metrics and data visualisation

Semester 2 2026

Subject Coordinator: **Dr Ekaterina Vylomova**

Co-lecturer: **Dr Hanzhuo (Vivian) Ma**

THE UNIVERSITY OF MELBOURNE

Using TF-IDF instead of Frequency

TF scores

3x128

	ambience	apartment	art	bathroom	bedroom	building	ceiling	city	downstairs	garden	house	inside
doc1	1	0	1	0	0	0	1	0	1	2	0	2
doc2	0	2	0	1	0	2						
doc3	0	0	0	2	2	0						

doc3 vs doc1: 0.41

doc3 vs doc2: 0.43

TF-IDF scores

	ambience	apartment	art	bathroom	bedroom	building	ceiling	city	downstairs
doc1	**0.1377**	0.0000	**0.1377**	0.0000	0.0000				
doc2	0.0000	**0.2419**	0.0000	**0.0920**	0.0000				
doc3	0.0000	0.0000	0.0000	**0.1355**	**0.1782**				

doc3 vs doc1: 0.27

doc3 vs doc2: 0.26

Using TF-IDF instead of Frequency

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3x128

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doc3	0	0	0	2	2	0						

doc3 vs doc1: 0.41
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doc3 vs doc1: 0.27
doc3 vs doc2: 0.26

TF-IDF scores; no stop-words

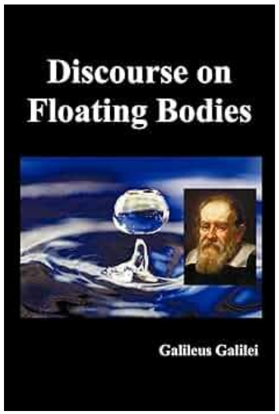
	ambience	apartment	art	bathroom	bedroom	building	ceiling	city	d
doc1	**0.1771**	0.0000	**0.1771**	0.0000	0.0000	0.0000	**0.1347**	0.0000	
doc2	0.0000	**0.2895**	0.0000	**0.1101**					
doc3	0.0000	0.0000	0.0000	**0.2229**					

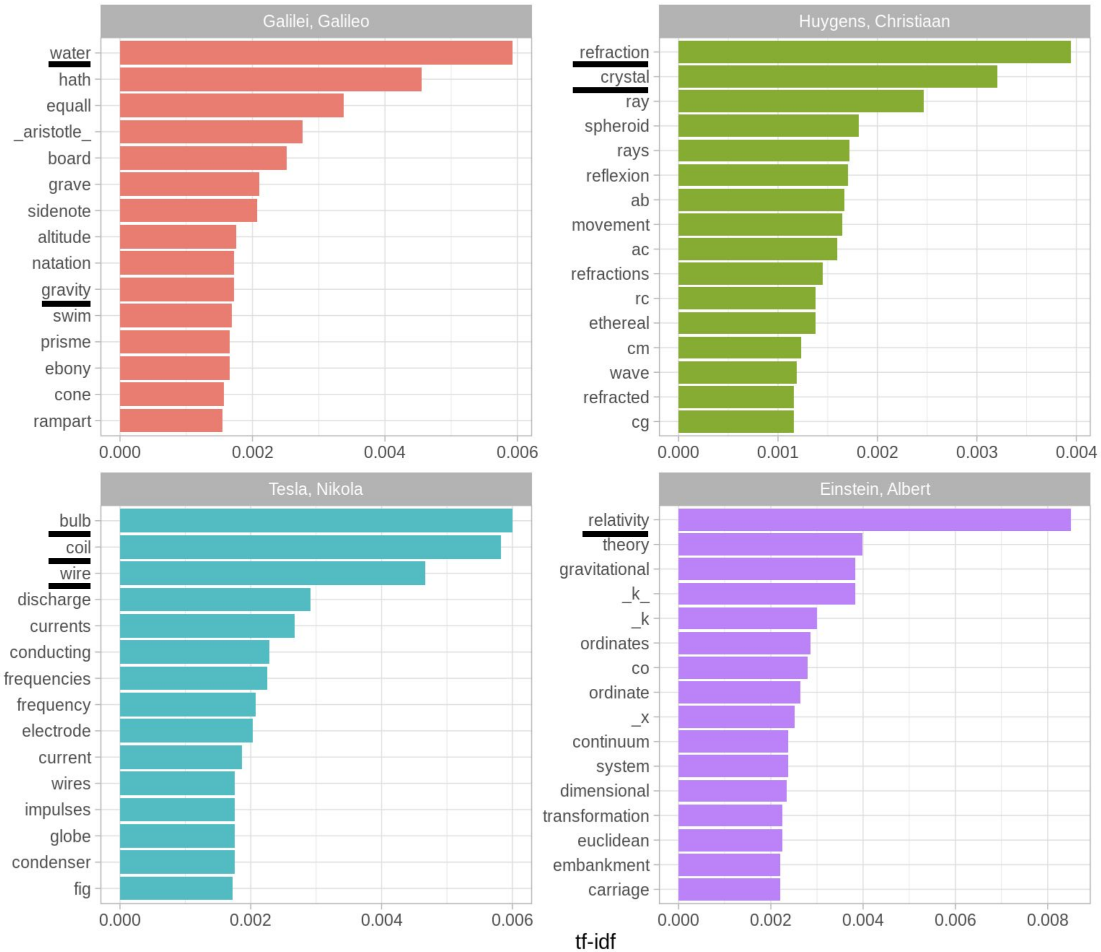
3x95

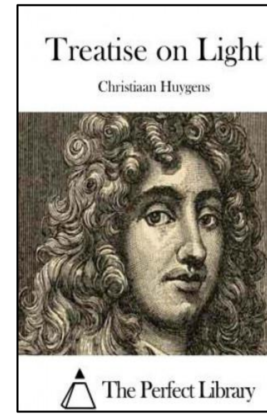
doc3 vs doc1: 0.16
doc3 vs doc2: 0.09

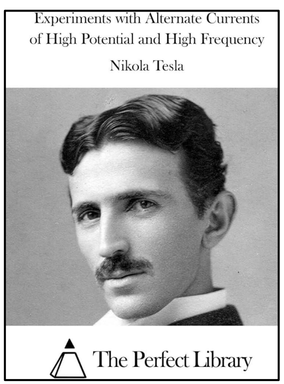
TFIDF Examples #2

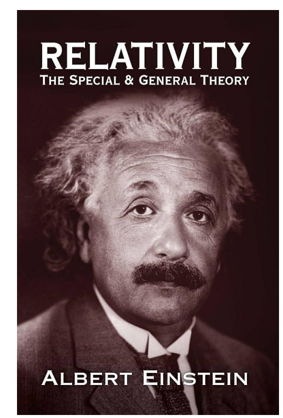
These are **NOT JUST frequent words**; they are **distinctive to each document**.









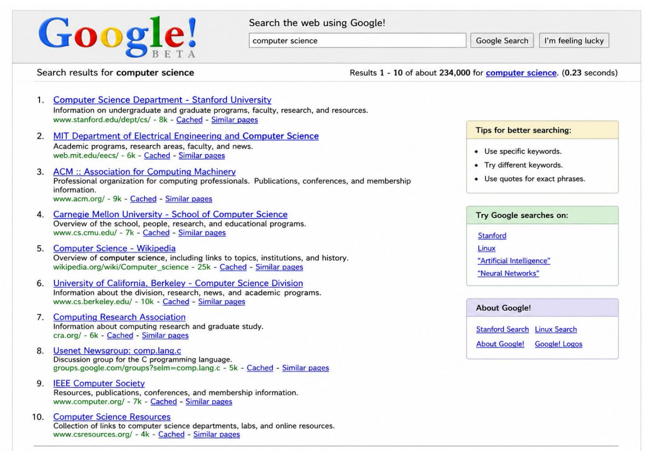


Why Document Similarity?

If we could calculate the similarity between any two documents, regardless of length, we could:

1. Arrange them into groups
2. Use one document to find similar ones
3. Put all the documents in a collection into one big 'space' and arrange them by similarity in that space
4. Use this to find the ones we want, given some important search words

Eventually, we could build



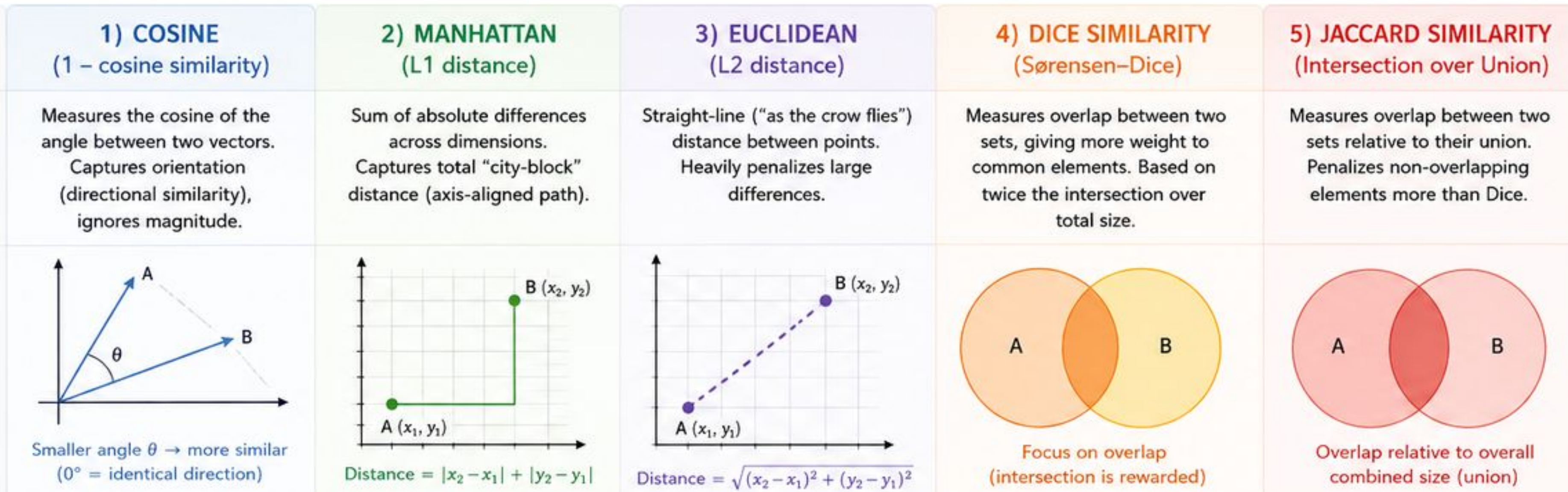
We need a distance metric!

To do anything useful in this space, you have to know how far one thing is from another

- A **distance metric** is a formula that gives you a score reflecting the distance between two points (vectors)

- We will learn a few of the common ones

The many forms of “distance”



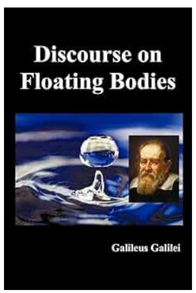
These are most commonly used only, there are many more…

Documents as vectors

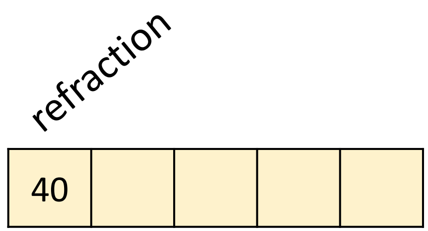
Let’s get familiar with using vectors to represent things

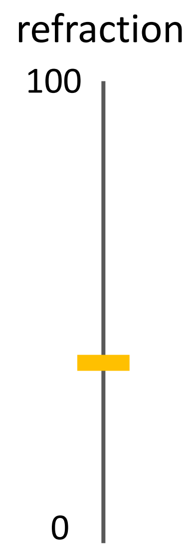
Assume that you are taking five known words to distinguish books:

1) refraction, 2) motion, 3) light, 4) electricity, 5) gravity



Galilei





Enjoy: https://www.gutenberg.org/files/37729/37729-h/37729-h.htm

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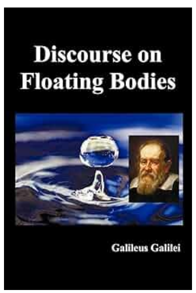
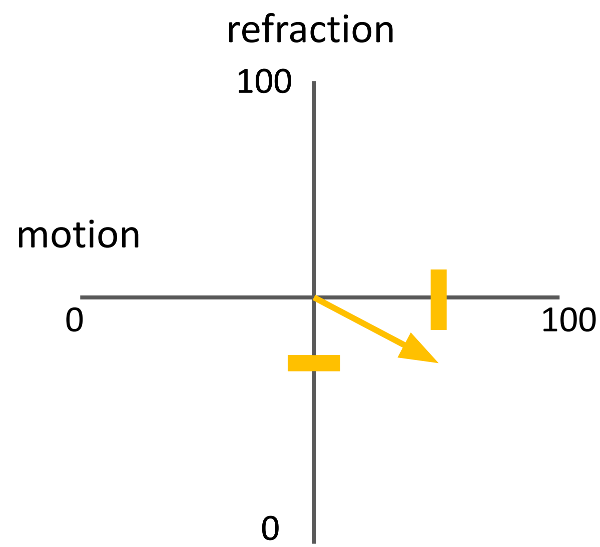


Galilei

refraction	motion			
40	70			



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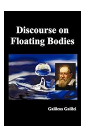
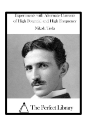
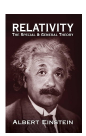
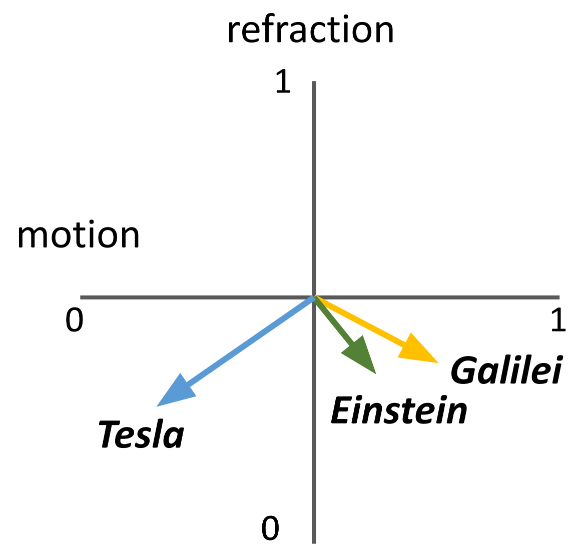






	refraction	motion			
Galilei	0.4	0.7			
Tesla	0.3	0.2			
Einstein	0.4	0.6			



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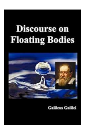
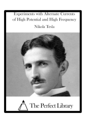
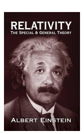
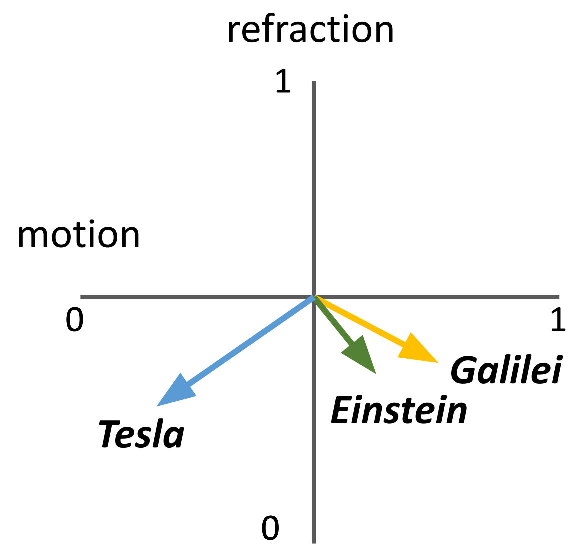
		refraction	motion			
	***Galilei***	0.4	0.7			
	***Tesla***	0.3	0.2			
	***Einstein***	0.4	0.6			

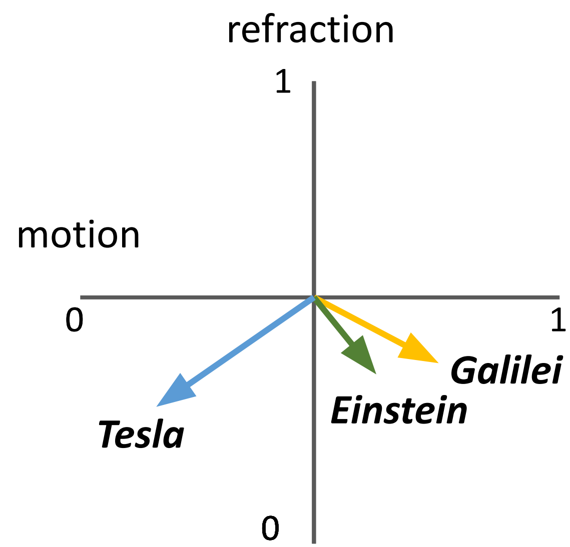


this dimension has higher variance → is more informative to distinguish the samples (related to PCA)

Cosine Distance (1-Similarity)

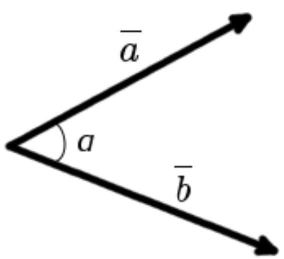
Let’s get familiar with using vectors to represent things

Which of the two documents (Tesla or Einstein) is more similar to Galilei?



Cosine Similarity

cosine of the angle between the two vectors



$$\cos \alpha = \frac{\bar{a} \cdot \bar{b}}{|a| \cdot |b|}$$

$$\bar{a} \cdot \bar{b} = a_x \cdot b_x + a_y \cdot b_y = 0.4 \cdot 0.3 + 0.7 \cdot 0.2 = 0.12 + 0.14 = 0.26$$

$$|\bar{a}| = \sqrt{a_x{}^2 + a_y{}^2} = \sqrt{0.4^2 + 0.7^2} = \sqrt{0.16 + 0.49} = \sqrt{0.65} = 0.1 \cdot \sqrt{65}$$

$$|\bar{b}| = \sqrt{b_x{}^2 + b_y{}^2} = \sqrt{0.3^2 + 0.2^2} = \sqrt{0.09 + 0.04} = \sqrt{0.13} = 0.1 \cdot \sqrt{13}$$

cos(*Galilei* , *Tesla*) ≈ 0.89

$$\cos \alpha = \frac{0.26}{0.1 \cdot \sqrt{65} \cdot 0.1 \cdot \sqrt{13}} = 0.4 \cdot \sqrt{5} \approx 0.8944271909999159$$

Cosine Similarity

Let’s get familiar with using vectors to represent things

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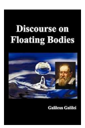
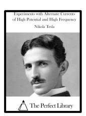
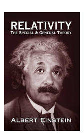
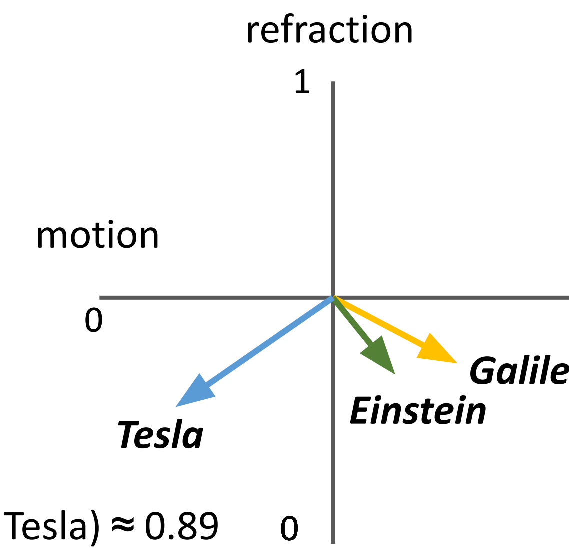
		refraction	motion			
	***Galilei***	0.4	0.7			
	***Tesla***	0.3	0.2			
	***Einstein***	0.4	0.6			



cos (Gal, Tesla) ≈ 0.89
cos (Gal, Einstein) ≈ 0.99
cos (Tesla, Einstein) ≈ 0.92

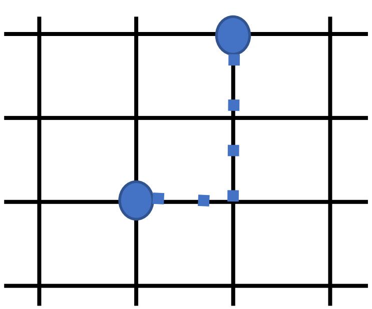
https://onlinemschool.com/math/assistance/vector/angl/
https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairwise.cosine_similarity.html

Manhattan Distance

Called **City Block distance** or
L1 distance (norm)

- Not a straight line between two points:
  you have to walk along the grid
  (the streets)
- The sum of the absolute difference between each dimension of 2 points:

$$\textbf{\textit{Manhattan}}\ [(x_1, y_1), (x_2, y_2)] = |x_1-x_2| + |y_1-y_2|$$



1+2=3

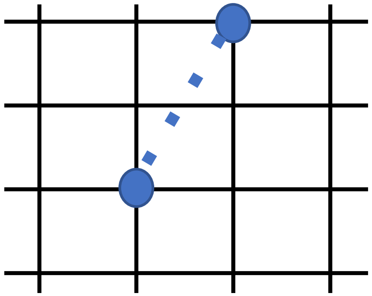
Euclidean Distance

The normal '**straight line' distance** on a flat plane

Also called "L2 distance"

- In 2 dimensions:

$$\textit{Euclidean}\,[(x_1, y_1), (x_2, y_2)] = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$



$$\sqrt{1^2+2^2} = 2.23606$$

Dice and Jaccard Similarity

Jaccard is used when you're looking at the data points (words) as sets:

$$\textbf{\textit{Jaccard similarity}}\ \mathbf{(A,B)} = |A \cap B| \ / \ |A \cup B|$$

$$\textbf{\textit{Jaccard distance}}\ \mathbf{(A,B)} = 1 - \text{Jaccard similarity}$$

A = {cat, dog}
B = {cat, dog, elephant}

Another popular metric is **Dice**:

$$\textbf{\textit{Dice similarity}}\ \mathbf{(A,B)} = 2|\ A \cap B\ |\ /\ (|A| + |B|)$$

More Distance metrics

- There are many more metrics (e.g, KL divergence compare distributions)
- There is no 'best' metric
  (that's why people keep inventing new ones)
- For more, see the web, for example
  https://www.kdnuggets.com/2020/11/most-popular-distance-metrics-knn.html

What can you use Distance for?

1. **Retrieval**: Finding a document like the one you have
2. **Duplicates**: Checking if you have more than one copy (or near-duplicates) of a document
3. **Clustering**: Finding sets of similar documents (e.g. news article clustering – sports, politics, etc.)
4. **Topic Modeling**: Finding topical themes in the data (e.g. “government policy affects the stock market” → finance topic 60%, politics topic 40%)
5. **Trend Analysis**: Seeing if document contents change over time (Distance comparison at time t vs t+1)

Vector Similarity

Remember?

Now, we want to view the data!

You can see **what you have**, **how it is balanced (or not)**, **what 'clusters' of data exist**, etc.

To Learn effective methods and tools to **encode quantitative and categorical information** so that is easy for others to *visually decode* the information.

- Visual methods to **find structures**
- Graphing/plotting is **a visual encoding of data**

Exploratory data analysis

Human eyes/brain are powerful to detect structures

Purpose of visualisation

- To analyse data – reveal patterns in the data
- To communicate data to others

Data-driven hypothesis <u>generation</u> vs **Formal hypothesis <u>testing</u>**

- Data-driven is ‘bottom-up’: the data —> you
- Hypothesis testing is ‘top-down’: your mind —> the data

What is a *distribution*?

- The values of a quantitative variable have a range (like: % is between zero and 100)
- Data points occupy positions in the range

The 'spread' or pattern of a set of data points is its *distribution*

- Simple statistics describe aspects of the distribution:
    - Measures of **location** (where are the numbers?)
    - Measures of **dispersion** (how are the numbers distributed?)

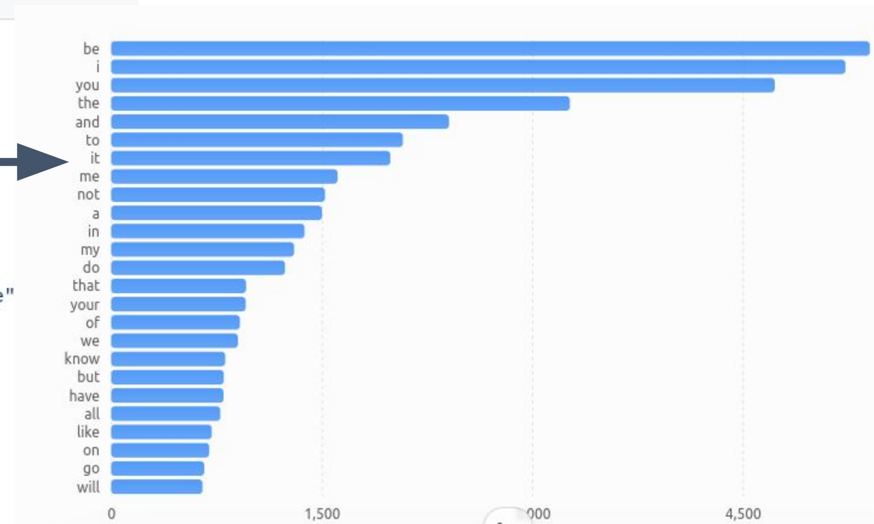
Examples

Corpus-of-Taylor-Swift / lyrics / flat-song-lyrics.json

sagesolar Fix lyric word 'cheer-captain' (#7)

Code Blame 12272 lines (12272 loc) · 701 KB


```
{
  "TSW:01:001:V": "He said the way my blue eyes shined",
  "TSW:01:002:V": "Put those Georgia stars to shame that night",
  "TSW:01:003:V": "I said, \"That's a lie\"",
  "TSW:01:004:V": "Just a boy in a Chevy truck",
  "TSW:01:005:V": "That had a tendency of gettin' stuck",
  "TSW:01:006:V": "On backroads at night",
  "TSW:01:007:C": "And I was right there beside him all summer long",
  "TSW:01:008:C": "And then the time we woke up to find that summer gone"
  "TSW:01:009:C": "But when you think Tim-McGraw",
  "TSW:01:010:C": "I hope you think my favorite song",
  "TSW:01:011:C": "The one we danced to all night long",
  "TSW:01:012:C": "The moon like a spotlight on the lake",
  "TSW:01:013:C": "When you think happiness",
  "TSW:01:014:C": "I hope you think that little black dress",
  "TSW:01:015:C": "Think of my head on your chest",
  "TSW:01:016:C": "And my old faded blue jeans",
  "TSW:01:017:C": "When you think Tim-McGraw",
  "TSW:01:018:C": "I hope you think of me",
  "TSW:01:019:V": "September saw a month of tears",
  "TSW:01:020:V": "And thankin' god that you weren't here",
  "TSW:01:021:V": "To see me like that",
  "TSW:01:022:V": "But in a box beneath my bed",
```

word frequency distribution (cut at 25 most frequent)

Examples


CAFE ORDERS
MONDAY
Alice
ordered two large cappuccinos and a blueberry muffin.
She paid $13.50.
2
Large Cappuccino
$5.00 each
1
Blueberry Muffin
$3.50
TOTAL PAID
$13.50
Bob
bought a flat white.
1
Flat White
$4.50
TOTAL PAID
$4.50
TUESDAY
Alice
came back and ordered a large cappuccino and two croissants.
The total was $15.00.
1
Large Cappuccino
$5.00
2
Croissant
$5.00 each
TOTAL PAID
$15.00
Charlie
ordered tea, but we forgot to record the price.
1
Tea
TOTAL PAID
?
(price not recorded)

Examples

20 drinks ordered: Latte, Flat white, Latte, Cappuccino, Tea, Latte, Flat white, Long black, Latte, Tea, Cappuccino, Latte, Flat white, Latte, Long black, Latte, Cappuccino, Tea, Latte, Flat white

Drink	Frequency
Latte	8
Flat white	4
Cappuccino	3
Tea	3
Long black	2

Examples

20 drinks ordered: Latte, Flat white, Latte, Cappuccino, Tea, Latte, Flat white, Long black, Latte, Tea, Cappuccino, Latte, Flat white, Latte, Long black, Latte, Cappuccino, Tea, Latte, Flat white

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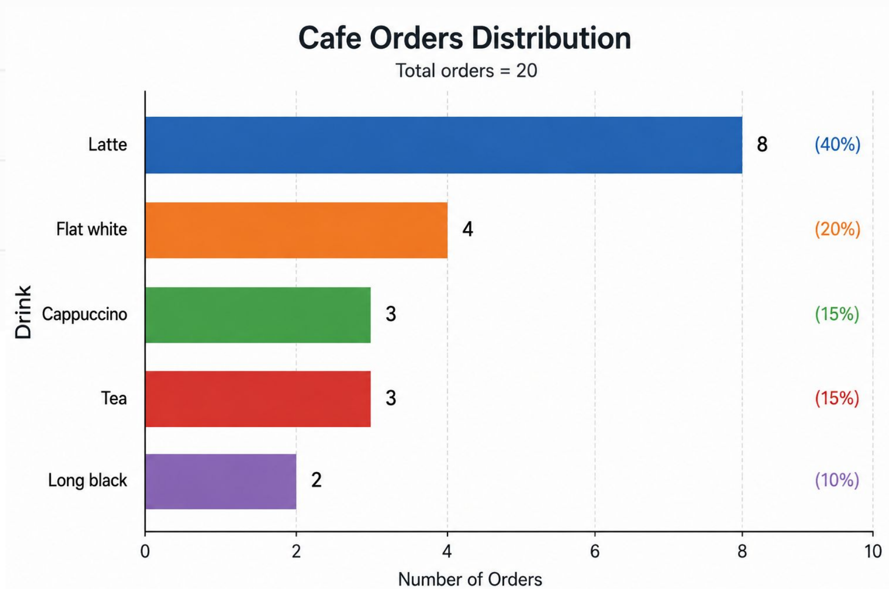


Measures of *location*

Location: **where** in the distribution is a useful indicator value?

- **Min**, **Max**: the smallest and largest values
- **Mean (average)**
  - Simple numeric
- **Median**: the middle value
  - For both Ordinal and Nominal variables
- **Mode**: the most common value
  - Numeric, Ordinal, and Nominal variables
- **Quartile**, **Percentile**: the quarter point or the percentage below

1, 2, 2, 2, 2, 3, 9

Mean = 21/7 = 3

Median

Mode 2

My son Mark is in the 99th percentile by height

Measures of *dispersion*

Dispersion: what is the **variability** of the distribution?

Variance

- 'wiggliness' – how far are the points from the mean/true value?

Standard Deviation

- kind of average variance

Range

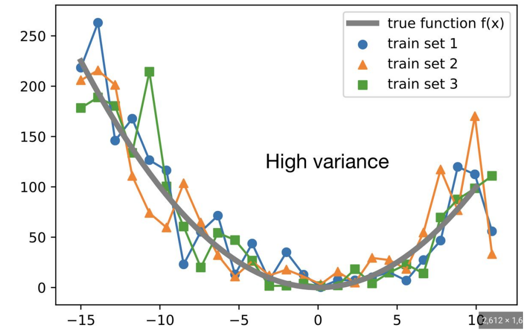
- from min to max

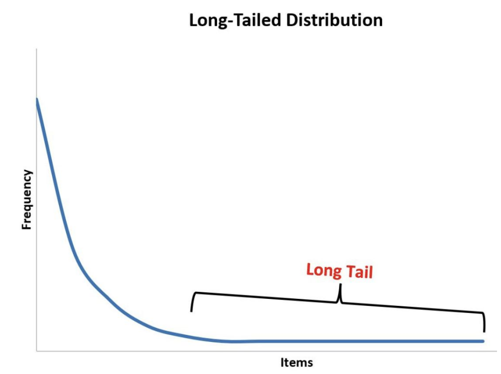
Interquartile range, **quantiles**

- where they lie

Skewness

- does a distribution have **a long tail?**





The formulas, in case you want them

Comparison of common averages of values [1, 2, 2, 3, 4, 7, 9]

Type	Description	Example	Result
Midrange	Midway point between the minimum and the maximum of a data set	**1**, 2, 2, 3, 4, 7, **9**	**5**
Arithmetic mean	Sum of values of a data set divided by number of values: $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$	(1 + 2 + 2 + 3 + 4 + 7 + 9) / 7	**4**
Median	Middle value separating the greater and lesser halves of a data set	1, 2, 2, **3**, 4, 7, 9	**3**
Mode	Most frequent value in a data set	1, **2**, **2**, 3, 4, 7, 9	**2**

$$2,\ 4,\ 4,\ 4,\ 5,\ 5,\ 7,\ 9.$$

These eight data points have the mean (average) of 5:

$$\mu = \frac{2+4+4+4+5+5+7+9}{8} = \frac{40}{8} = 5.$$

First, calculate the deviations of each data point from the mean, and square the result of each:

$$\begin{array}{ll} (2-5)^2 = (-3)^2 = 9 & (5-5)^2 = 0^2 = 0 \\ (4-5)^2 = (-1)^2 = 1 & (5-5)^2 = 0^2 = 0 \\ (4-5)^2 = (-1)^2 = 1 & (7-5)^2 = 2^2 = 4 \\ (4-5)^2 = (-1)^2 = 1 & (9-5)^2 = 4^2 = 16. \end{array}$$

The variance is the mean of these values:

Variance

$$\sigma^2 = \frac{9+1+1+1+0+0+4+16}{8} = \frac{32}{8} = 4.$$

and the *population* standard deviation is equal to the square root of the variance:

Standard deviation

$$\sigma = \sqrt{4} = 2.$$

Producing descriptive statistics

pandas.DataFrame.describe

`DataFrame.describe(percentiles=None, include=None, exclude=None, datetime_is_numeric=False)`

Generate descriptive statistics.

Describing a numeric `Series`.

```
>>> s = pd.Series([1, 2, 3])
>>> s.describe()
count    3.0
mean     2.0
std      1.0
min      1.0
25%      1.5
50%      2.0
75%      2.5
max      3.0
dtype: float64
```

Describing a categorical `Series`.

```
>>> s = pd.Series(["a", "a", "b", "c"])
>>> s.describe()
count      4
unique     3
top        a
freq       2
dtype: object
```

Describing a `DataFrame`. By default only numeric fields are returned.

```
>>> df = pd.DataFrame(
...     {
...         "categorical": pd.Categorical(["d", "e", "f"]),
...         "numeric": [1, 2, 3],
...         "object": ["a", "b", "c"],
...     }
... )
>>> df.describe()
      numeric
count     3.0
mean      2.0
std       1.0
min       1.0
25%       1.5
50%       2.0
75%       2.5
max       3.0
```

Describing all columns of a `DataFrame` regardless of data type.

```
>>> df.describe(include="all")
       categorical  numeric object
count            3      3.0      3
unique           3      NaN      3
top              f      NaN      a
freq             1      NaN      1
mean           NaN      2.0    NaN
std            NaN      1.0    NaN
min            NaN      1.0    NaN
25%            NaN      1.5    NaN
50%            NaN      2.0    NaN
75%            NaN      2.5    NaN
max            NaN      3.0    NaN
```

Univariate data:
Exploring single variables

Displaying single variables

- Histograms
- Outliers
- Binning
- Boxplots

Example data

https://datarepository.wolframcloud.com/resources/Sample-Data-Singer-Heights

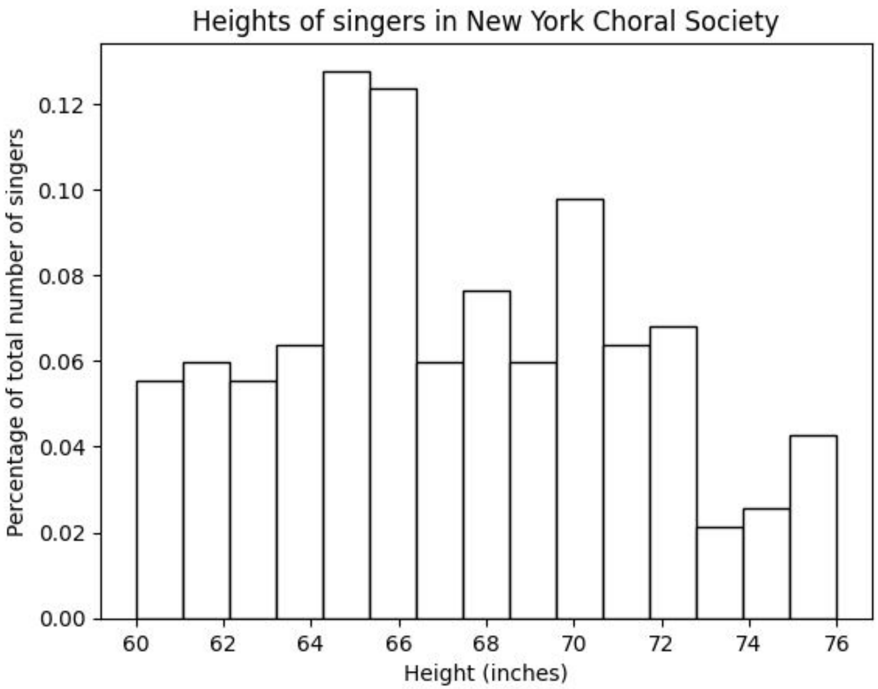
- Heights of 235 singers in the New York Choral Society
- Singers divided into 8 voice parts
  - Bass 1, Bass 2
  - Tenor 1, Tenor 2
  - Alto 1, Alto 2,
  - Soprano 1, Soprano 2
- The distribution of heights
- Differences among voice parts


```
"Height","Class"
"Quantity[64, ""Inches""]","Soprano1"
"Quantity[62, ""Inches""]","Soprano1"
"Quantity[66, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[60, ""Inches""]","Soprano1"
"Quantity[61, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[66, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[63, ""Inches""]","Soprano1"
"Quantity[67, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[62, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[68, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[63, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[62, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[66, ""Inches""]","Soprano1"
"Quantity[62, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[63, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[66, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[62, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[66, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[61, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[66, ""Inches""]","Soprano1"
"Quantity[65, ""Inches""]","Soprano1"
"Quantity[62, ""Inches""]","Soprano1"
```

Histograms

x-axis:
divide the range of values into consecutive, non-overlapping, and equal width intervals

y-axis:
frequency or values proportional to the frequency or count



pandas.DataFrame.hist

```
DataFrame.hist(column=None, by=None, grid=True, xlabelsize=None,
xrot=None, ylabelsize=None, yrot=None, ax=None, sharex=False,
sharey=False, figsize=None, layout=None, bins=10, backend=None,
legend=False, **kwargs)
```

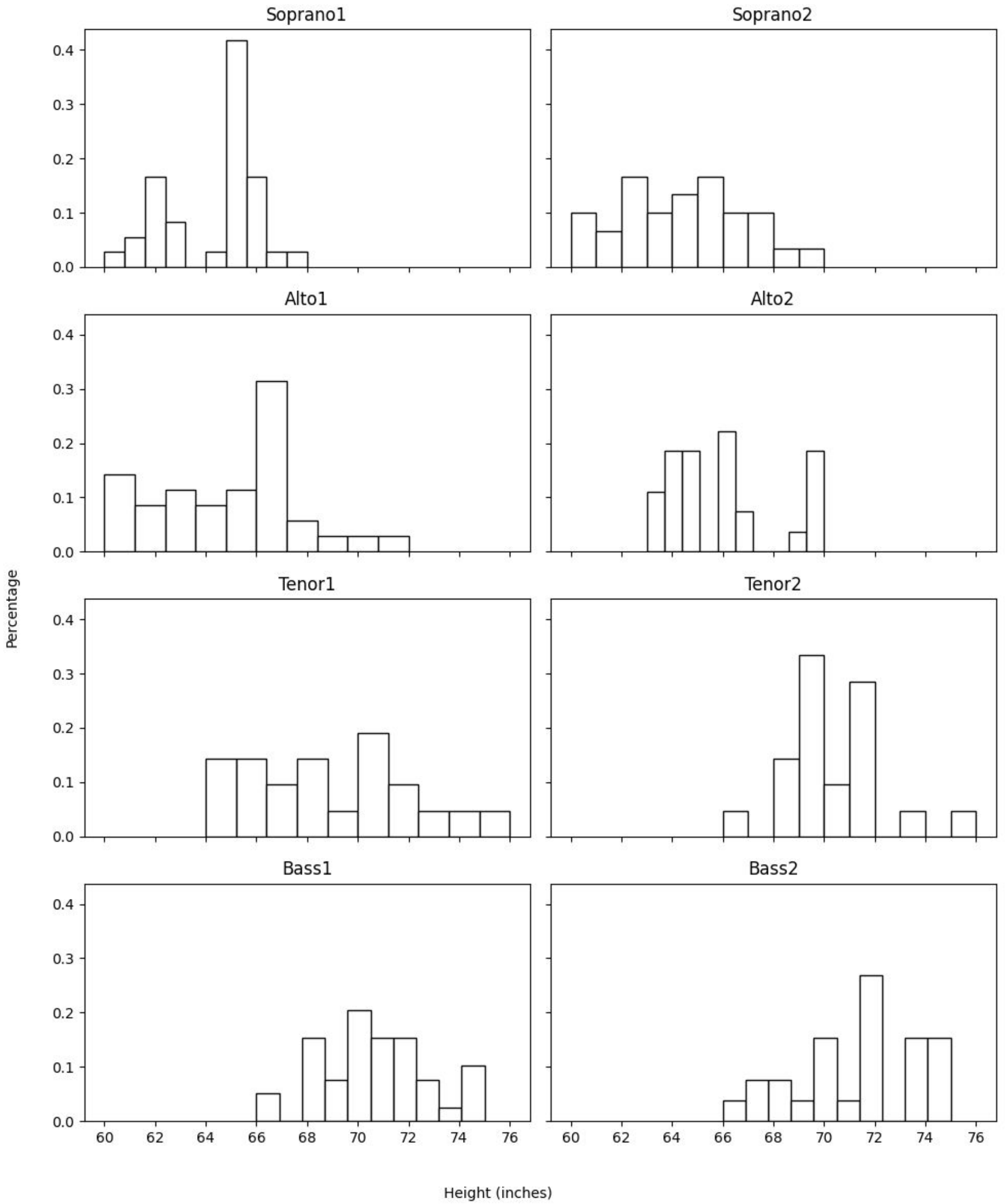
Make a histogram of the DataFrame's columns.

1 inch = 2.54 cm

Histograms

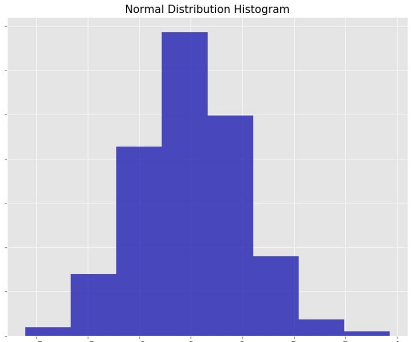
- Range
- Central location
- Symmetry
- Modality

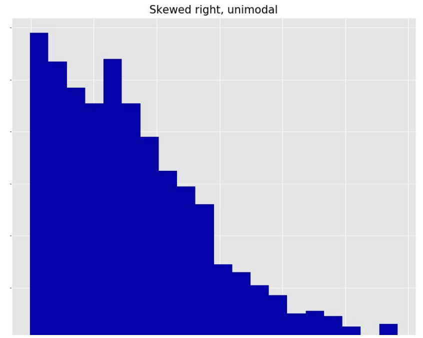
- Outliers

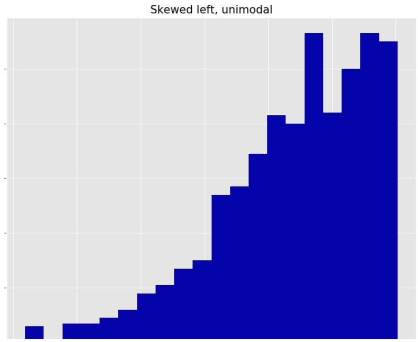


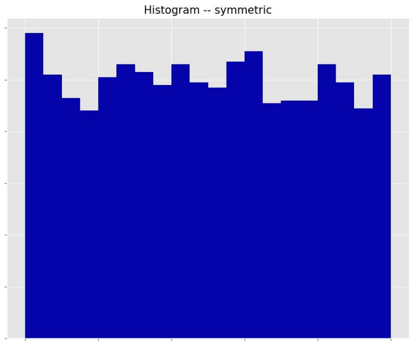
Histogram patterns

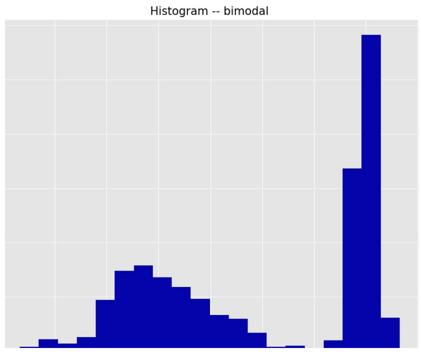
Symmetric, left/right skewed, unimodal, bimodal, multimodal

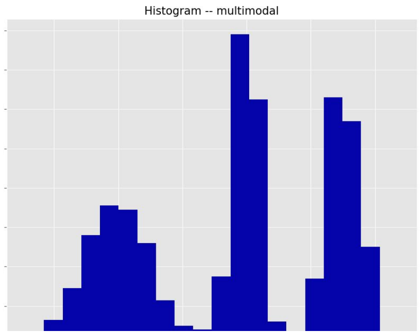






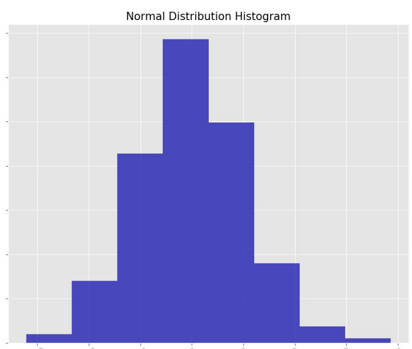


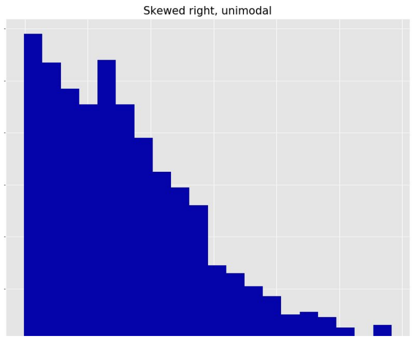


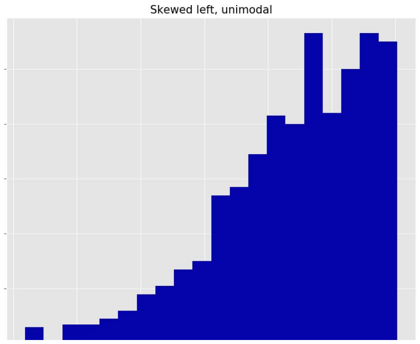


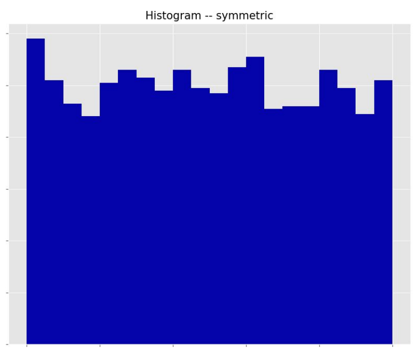
Histogram patterns

Symmetric, left/right skewed, unimodal, bimodal, multimodal

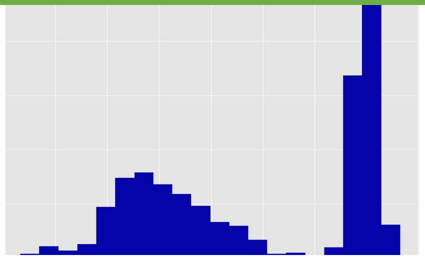


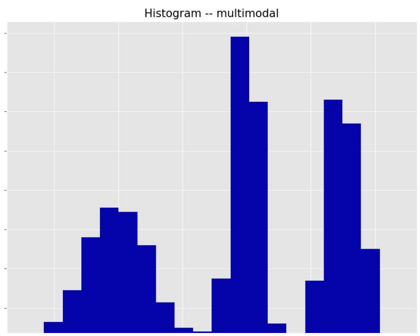






Very common in student grades





Outliers

- Gestation histogram:

  Historical case: The husband sued for divorce on the grounds of adultery, stating he had been away on military service 349 days before the birth

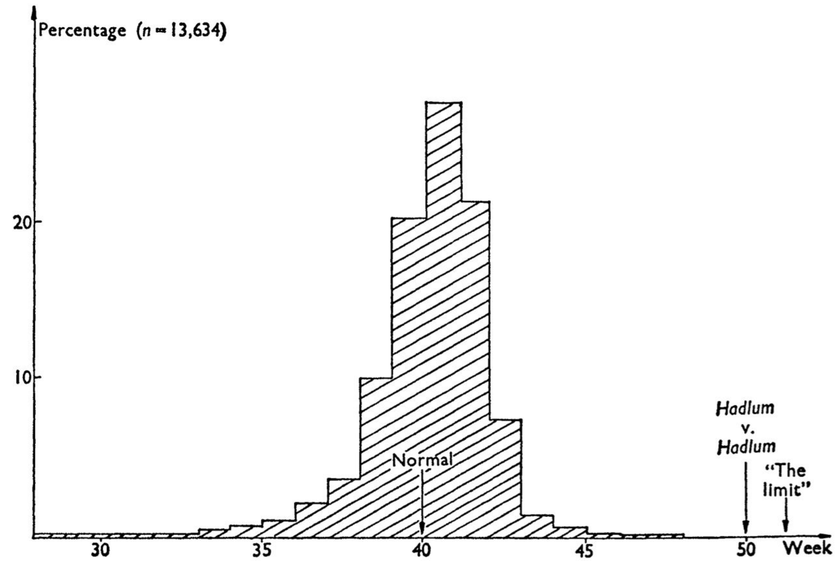
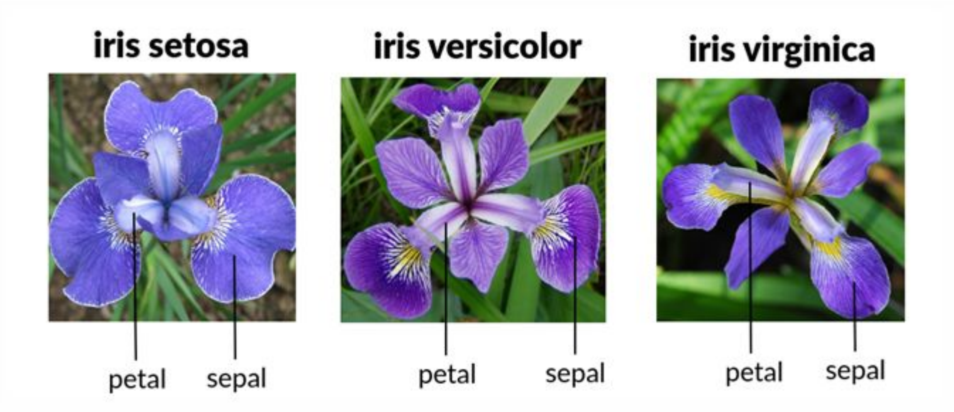


FIG. 1. Distribution of human gestation periods.

- Sure, the gestation was an outlier
- But was it 'too long'?
- A convenient definition of an **outlier**: a point that falls more than 1.5 times the interquartile range above the third quartile (or below the first quartile)

Binning 1: Iris dataset example

- Well known dataset introduced by statistician Ronald Fisher with 150 objects (https://en.wikipedia.org/wiki/Iris_flower_data_set)
- Four features
  - sepal length in cm
  - sepal width in cm
  - petal length in cm
  - petal width in cm
- Three flower species (classes):
  - Setosa
  - Virginica
  - Versicolour



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  - Versicolour

```
>>> from sklearn.datasets import load_iris
>>>
>>> iris = load_iris()
>>> print(iris)
{'data': array([[5.1, 3.5, 1.4, 0.2],
       [4.9, 3. , 1.4, 0.2],
       [4.7, 3.2, 1.3, 0.2],
       [4.6, 3.1, 1.5, 0.2],
       [5. , 3.6, 1.4, 0.2],
       [5.4, 3.9, 1.7, 0.4],
       [4.6, 3.4, 1.4, 0.3],
       [5. , 3.4, 1.5, 0.2],
       [4.4, 2.9, 1.4, 0.2],
       [4.9, 3.1, 1.5, 0.1],
       [5.4, 3.7, 1.5, 0.2],
       [4.8, 3.4, 1.6, 0.2],
       [4.8, 3. , 1.4, 0.1]
```

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- Four features
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       [4.9, 3. , 1.4, 0.2],
       [4.7, 3.2, 1.3, 0.2],
       [4.6, 3.1, 1.5, 0.2],
       [5. , 3.6, 1.4, 0.2],
       [5.4, 3.9, 1.7, 0.4],
       [4.6, 3.4, 1.4, 0.3],
       [5. , 3.4, 1.5, 0.2],
       [4.4, 2.9, 1.4, 0.2],
       [4.9, 3.1, 1.5, 0.1],
       [5.4, 3.7, 1.5, 0.2],
       [4.8, 3.4, 1.6, 0.2],
       [4.8, 3. , 1.4, 0.1]
```

Binning 1: Iris dataset example

Petal width:

[0.2, 0.2, 0.2, 0.2, 0.2, 0.4, 0.3, 0.2, 0.2, 0.1, 0.2, 0.2, 0.1, 0.1, 0.2, 0.4, 0.4, 0.3, 0.3, 0.3, 0.2, 0.4, 0.2, 0.5, 0.2, 0.2, 0.4, 0.2, 0.2, 0.2, 0.2, 0.4, 0.1, 0.2, 0.2, 0.2, 0.2, 0.1, 0.2, 0.2, 0.3, 0.3, 0.2, 0.6, 0.4, 0.3, 0.2, 0.2, 0.2, 0.2, 1.4, 1.5, 1.5, 1.3, 1.5, 1.3, 1.6, 1.0, 1.3, 1.4, 1.0, 1.5, 1.0, 1.4, 1.3, 1.4, 1.5, 1.0, 1.5, 1.1, 1.8, 1.3, 1.5, 1.2, 1.3, 1.4, 1.4, 1.7, 1.5, 1.0, 1.1, 1.0, 1.2, 1.6, 1.5, 1.6, 1.5, 1.3, 1.3, 1.3, 1.2, 1.4, 1.2, 1.0, 1.3, 1.2, 1.3, 1.3, 1.1, 1.3, 2.5, 1.9, 2.1, 1.8, 2.2, 2.1, 1.7, 1.8, 1.8, 2.5, 2.0, 1.9, 2.1, 2.0, 2.4, 2.3, 1.8, 2.2, 2.3, 1.5, 2.3, 2.0, 2.0, 1.8, 2.1, 1.8, 1.8, 1.8, 2.1, 1.6, 1.9, 2.0, 2.2, 1.5, 1.4, 2.3, 2.4, 1.8, 1.8, 2.1, 2.4, 2.3, 1.9, 2.3, 2.5, 2.3, 1.9, 2.0, 2.3, 1.8]

Binning 1: Iris dataset example

Petal width:

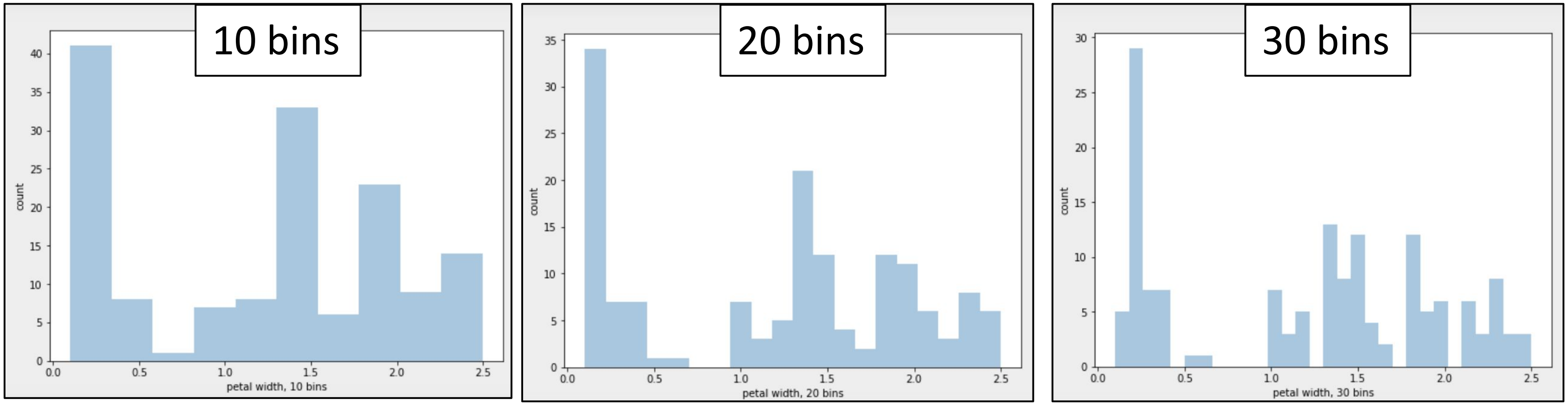
[0.1, 0.1, 0.1, 0.1, 0.1,
0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2,
0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.3,
0.4, 0.4, 0.4, 0.4, 0.4, 0.4, 0.4,
0.5,
0.6,
1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0,
1.1, 1.1, 1.1,
1.2, 1.2, 1.2, 1.2, 1.2,
1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3,
1.4, 1.4, 1.4, 1.4, 1.4, 1.4, 1.4, 1.4,
1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5,
1.6, 1.6, 1.6, 1.6,
1.7, 1.7,
1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8,
1.9, 1.9, 1.9, 1.9, 1.9,
2.0, 2.0, 2.0, 2.0, 2.0, 2.0,
2.1, 2.1, 2.1, 2.1, 2.1, 2.1,
2.2, 2.2, 2.2,
2.3, 2.3, 2.3, 2.3, 2.3, 2.3, 2.3, 2.3,
2.4, 2.4, 2.4,
2.5, 2.5, 2.5]

How to represent the distribution?

How to split the data into groups (bins)?

Binning 2: Histogram of petal width

Histograms of the same dataset may look different with different bin sizes:



Problem: **Hard to choose appropriate bin size** for histogram

- Too small → normal objects in empty/rare bins, false positive
- Too big → outliers in some frequent bins, false negative

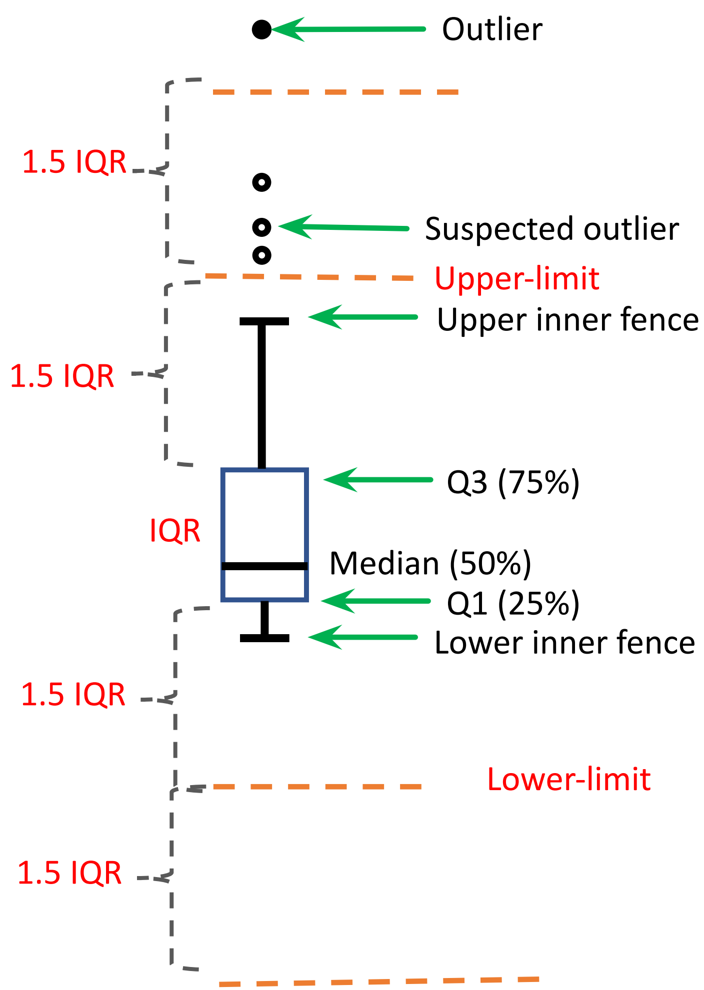
Tukey's box-and-whisker plot

- **Median**: the middle data point (once sorted)
- **Q1**: middle data point below the median
- **Q3**: middle data point above the median
- **IQR** = interquartile range = Q3 – Q1

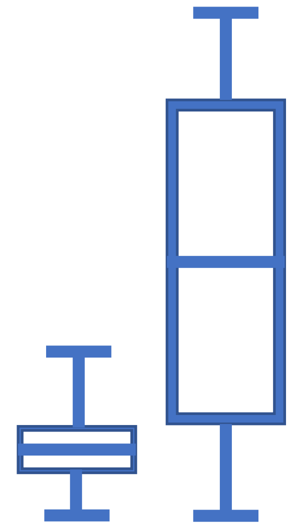
- **Whiskers** (inner fence)
  - Upper-limit: **Q3 + 1.5 × IQR**
  - Upper inner fence: Highest data point ≤ Upper-limit

- **Lower-limit** Q1 - 1.5 × IQR
- **Lower inner fence**: Lowest data point ≥ Lower-limit

- **Suspected outliers** (circle)
  - > 1.5 × QR below Q1 or above Q3
- **Outliers** (black dot)
  - > 3 × IQR below Q1 or above Q3

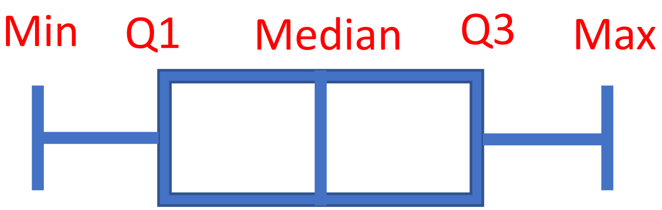


Box plot patterns



Symmetric or skewed?
Tightly or loosely grouped?

Symmetric



Left skew

Right skew

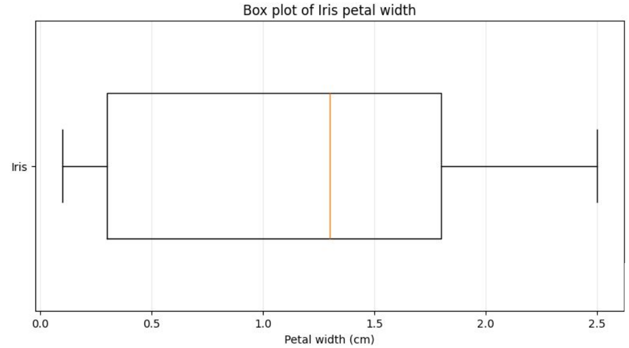
Example: Petal Width

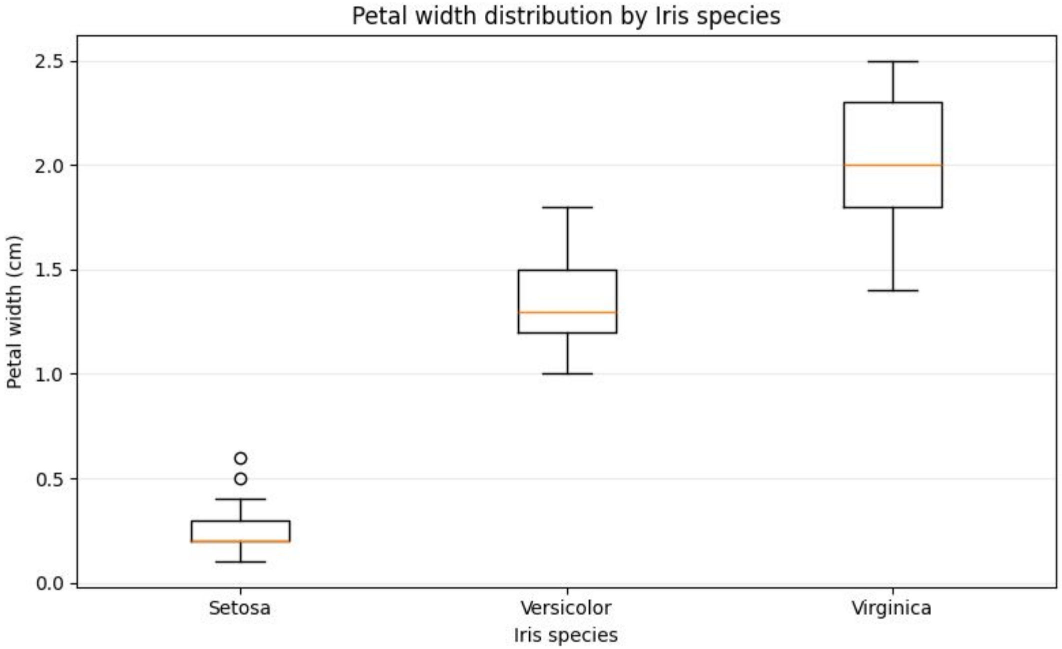
[0.1, 0.1, 0.1, 0.1, 0.1,
0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 0.2,
0.3, 0.3, 0.3,⭐ 0.3 ← Q1, 0.3, 0.3, 0.3,
0.4, 0.4, 0.4, 0.4, 0.4, 0.4, 0.4,
0.5,
0.6,
1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0,
1.1, 1.1, 1.1,
1.2, 1.2, 1.2, 1.2, 1.2,
1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, ⭐ 1.3 ← Median, 1.3, 1.3,
1.4, 1.4, 1.4, 1.4, 1.4, 1.4, 1.4, 1.4,
1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5,
1.6, 1.6, 1.6, 1.6,
1.7, 1.7,
1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, 1.8, ⭐ 1.8 ← Q3, 1.8, 1.8, 1.8,
1.9, 1.9, 1.9, 1.9, 1.9,
2.0, 2.0, 2.0, 2.0, 2.0, 2.0,
2.1, 2.1, 2.1, 2.1, 2.1, 2.1,
2.2, 2.2, 2.2,
2.3, 2.3, 2.3, 2.3, 2.3, 2.3, 2.3, 2.3,
2.4, 2.4, 2.4,
2.5, 2.5, 2.5]

Number of observations n	**150**
Minimum	**0.1**
First quartile Q_1	**0.3**
Median Q_2	**1.3**
Third quartile Q_3	**1.8**
Maximum	**2.5**
Interquartile range $IQR = Q_3 - Q_1$	**1.5**

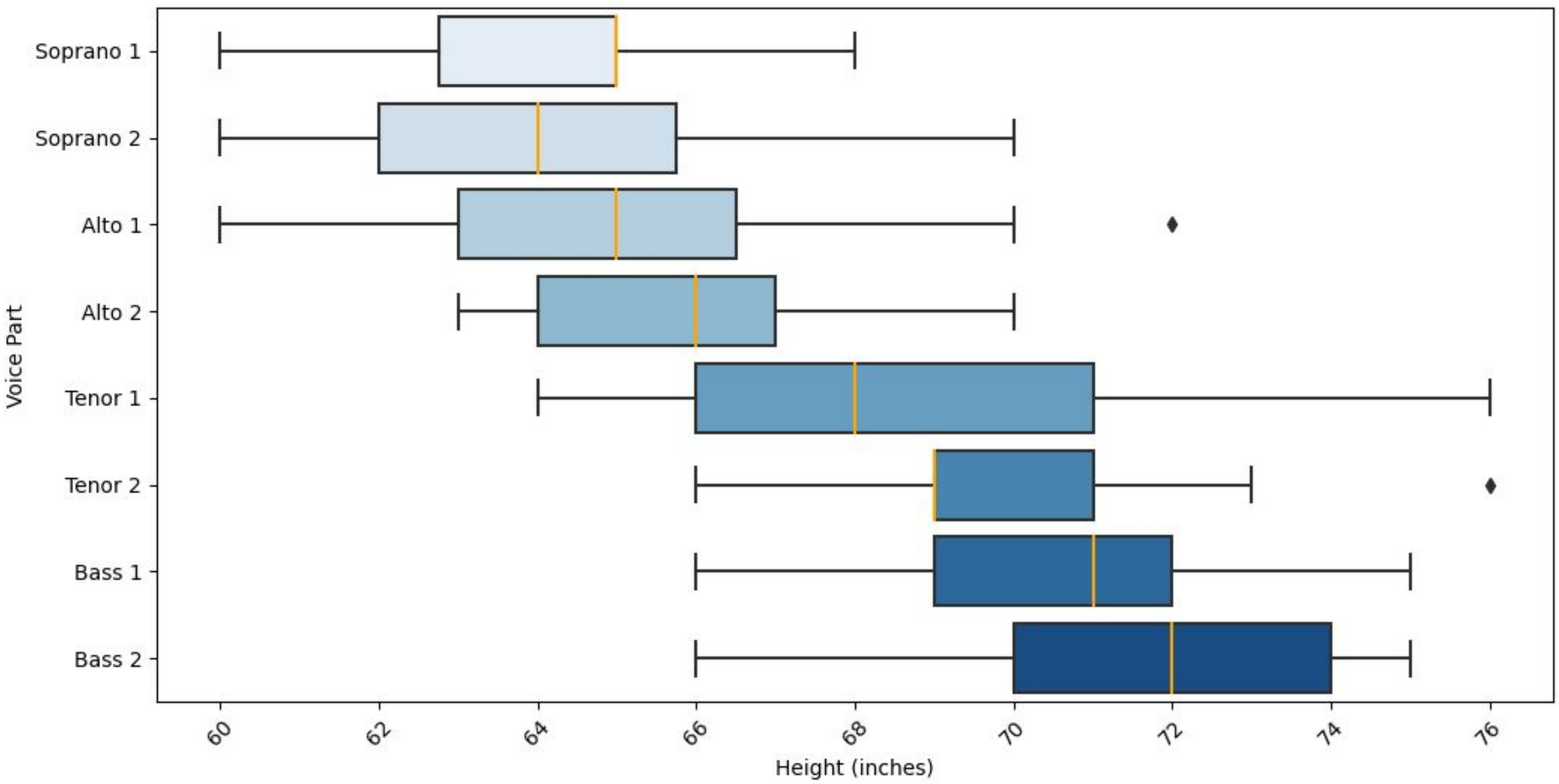
Example: Petal Width

All:





Comparing box plots



Bi- and trivariate data: Exploring pairs and triples of variables

Scatter plots 1

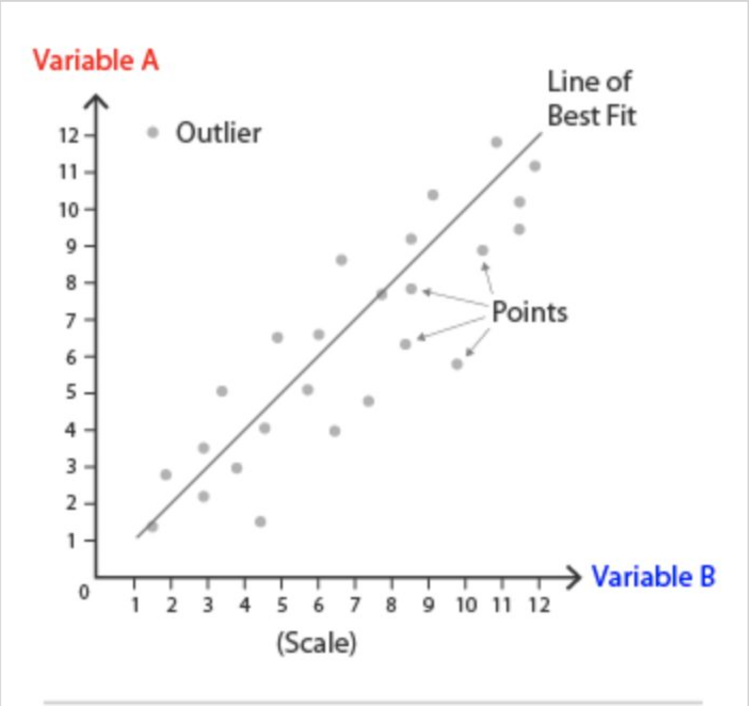
- display values for typically two variables for a set of data
- Example for flats: the **larger**, the more **expensive**

https://www.data-to-viz.com/graph/scatter.html

Scatter plots 2

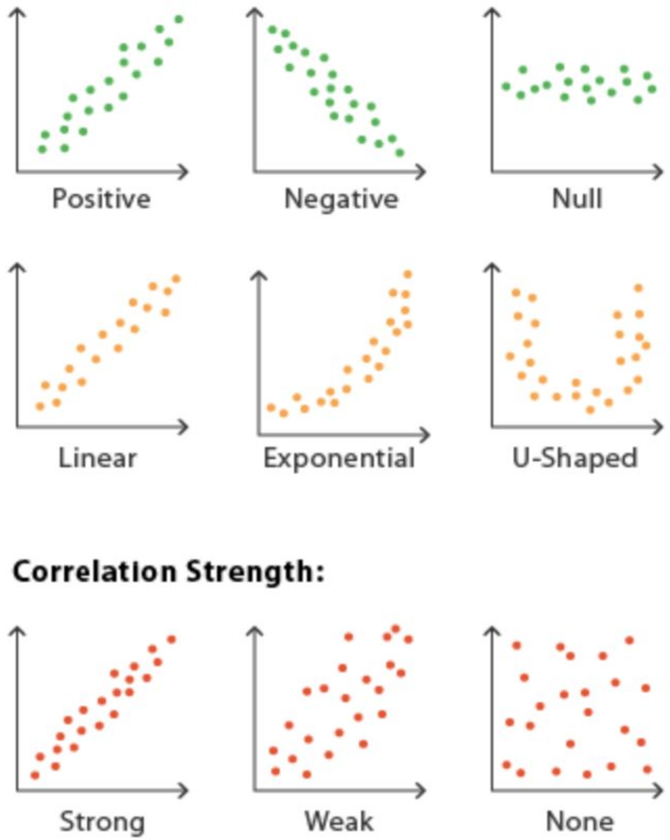
Two numeric variables

- **X-axis**: one numeric variable
- **Y-axis**: the other numeric variable
- A dot is a data point with 2 values as the x, y coordinates



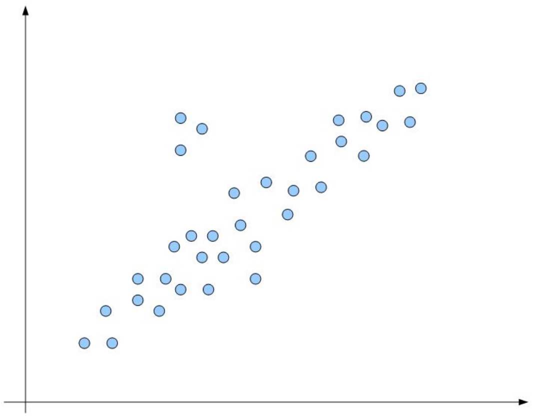
https://datavizcatalogue.com/methods/scatterplot.html

Scatter plots 3: Patterns and outliers



https://datavizcatalogue.com/methods/scatterplot.html

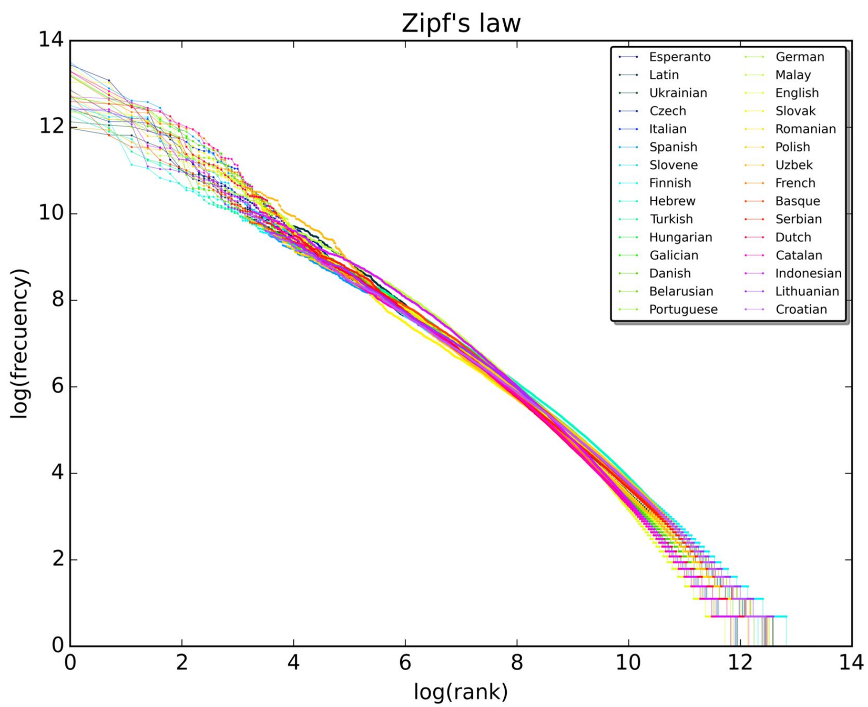
Typical correlation pattern: As x grows, so does y, usually



https://nirpyresearch.com/outliers-detection-pls-regression-nir-spectroscopy-python/

Scatter plots 3: Patterns and outliers

– **Zipf's Law** (word frequency distributions)

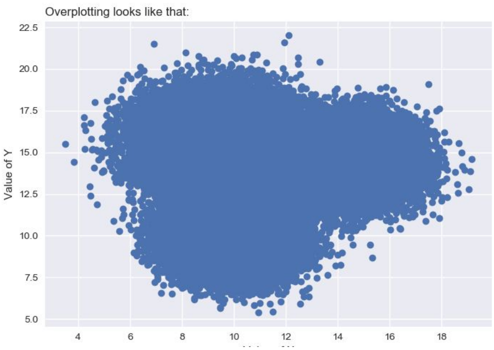


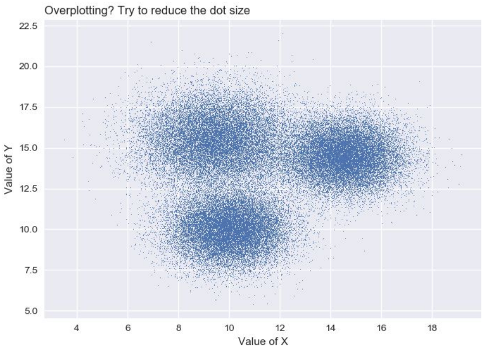
$$\text{word frequency} \propto \frac{1}{\text{word rank}}$$

‘Over plotting’ in scatter plots

When there are many data points, dots may overlap

- For moderate overplotting
  - Reduce dot size
  - Jittering: a small amount of uniform random noise is added to the values before graphing
- Sampling
- Use other plots **such as contour plots**



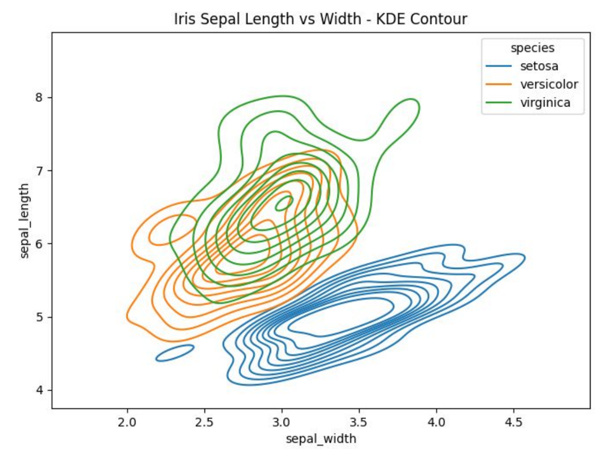


See https://python-graph-gallery.com/134-how-to-avoid-overplotting-with-python/

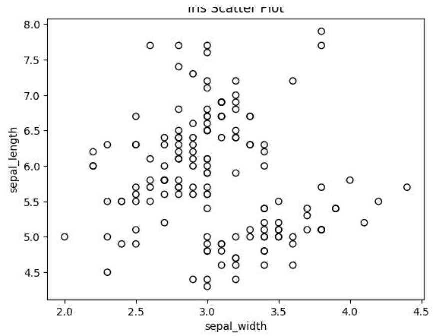
Contour plot 1

- **3 variables: x, y, and z**
- Still plot them in 2 dimensions, but can show more
- Contour lines: Level lines that **connect the points** where the values of the **3rd variable are the same**
- Can reveal if data are overplotted in scatter plots

- Iris data:



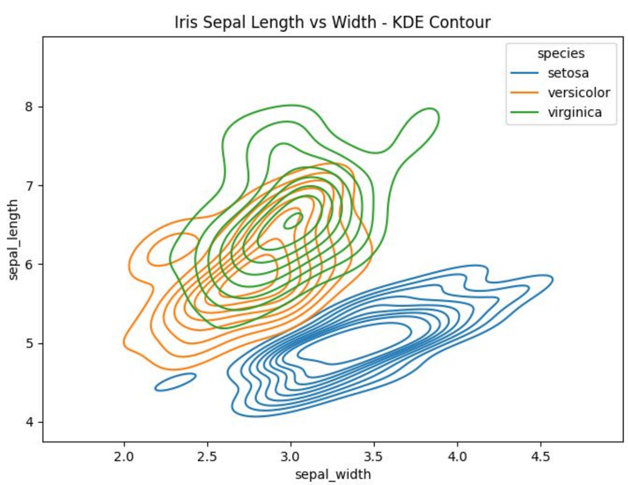
Contour plot 2



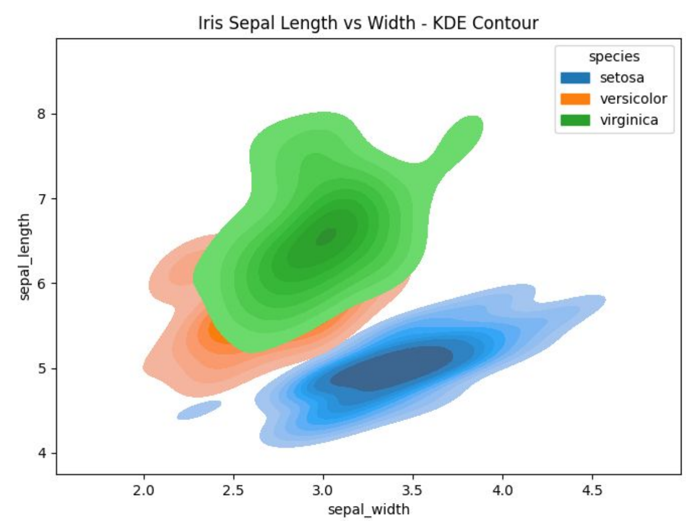
Python seaborn library

- sns.kdeplot()
- **density of data points** as the 3rd variable, calculated using **kernel density estimate**

`fill=False`



`fill=True`



High-dimensional data: Exploring more than 3 dimensions

More than 3 dimensions

- **Maps, tables, graphs**: we are used to **2 dimensions**

- When you need to show more values, you can use additional ‘info carriers’:
  - **Colour**
  - **Shape**
  - **Size**
  - **Movement**

- Use a Legend to define what each carrier shows

Iris data

Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.7	2.9	4.2	1.3	versicolour
6.2	2.9	4.3	1.3	versicolour
5.1	2.5	3.0	1.1	versicolour
5.7	2.8	4.1	1.3	versicolour
7.6	3.0	6.6	2.1	virginica
4.9	2.5	4.5	1.7	virginica
7.3	2.9	6.3	1.8	virginica
6.7	2.5	5.8	1.8	virginica

Which parameter would you use to determine the type of iris?

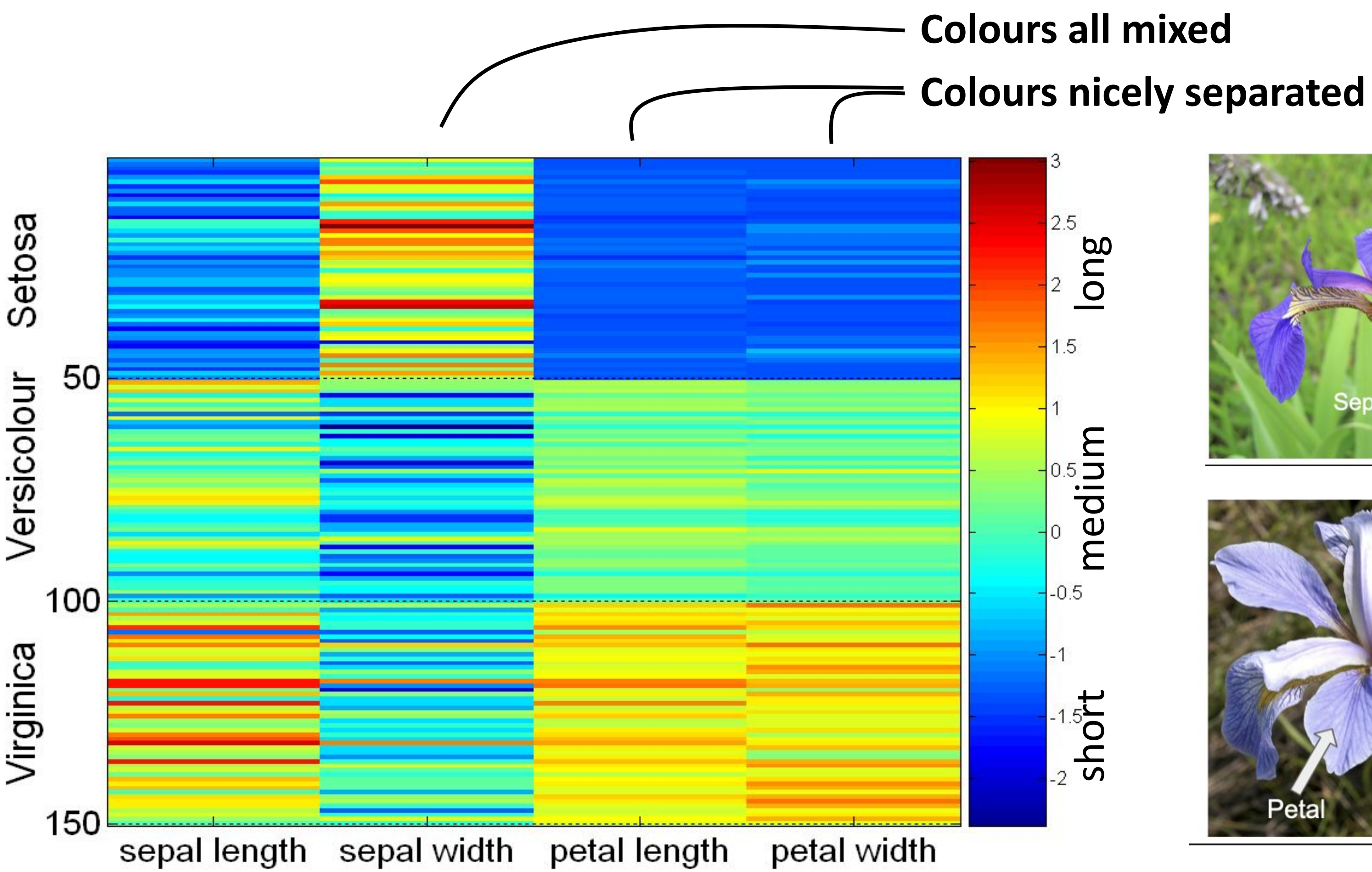
Clearly, **petal width**:
- short = setosa
- medium = versicolour
- long = virginica

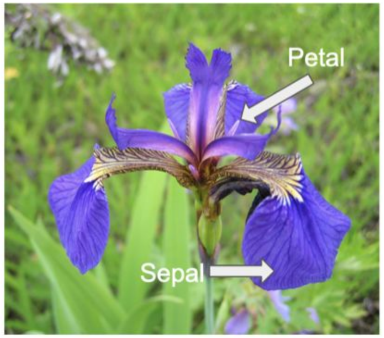
The other parameters don't distinguish cleanly

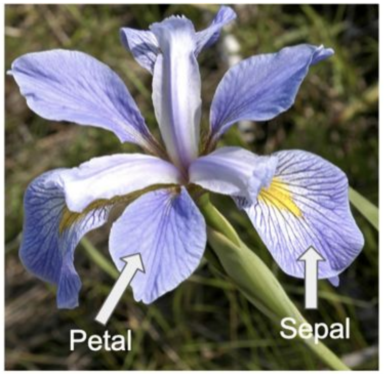
How can you determine this?

- You can infer the patterns because the table was nicely sorted or because you spent long hours poring over the data
- The computer can help you find the patterns by clever visualization
  - Heat map
  - Scatterplot matrix
  - PCA and clustering (next lecture)

Heat map of standardised iris data



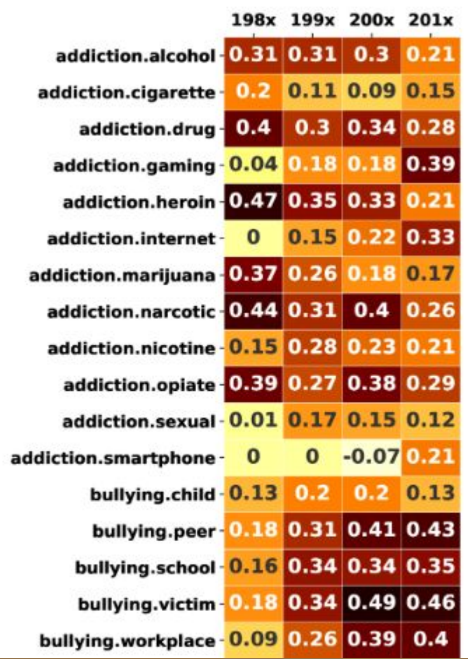




[Columns have been standardised to have a mean of zero and standard

Heat map considerations

- For 3-D data: show x and y axes, and display the **z values by colour**
- Plot the data matrix
- This can be useful when objects are sorted according to class/type
- Typically, features are normalised to prevent one attribute from dominating the plot



[Evaluation of Semantic Change of Harm-Related Concepts in Psychology](#) (Vylomova et al., 2019)

How common is your birthday?

The numbers in the squares rank birthdays from most common (1) to least common (366).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	363	97	210	294	61	252	36	120	189	4	326	342
2	297	172	72	18	158	233	130	301	121	128	321	334
3	269	146	92	154	264	63	170	218	50	183	333	304
4	59	37	8	32	213	41	213	211	196	280	258	332
5	76	16	10	276	119	81	291	88	185	274	157	348
6	162	116	83	331	56	147	196	79	289	206	281	355
7	71	262	34	104	93	276	200	153	254	131	300	350
8	52	143	260	2	165	295	68	125	171	25	329	347
9	255	141	124	143	165	323	90	263	48	63	321	339
10	272	106	127	230	271	114	218	216	33	136	292	286
11	140	46	24	59	193	95	244	216	250	265	226	316
12	163	6	26	175	159	109	269	90	110	246	179	327
13	259	243	209	320	222	317	265	195	302	237	313	357
14	72	242	152	151	43	330	239	188	83	76	311	351
15	110	78	223	28	199	126	122	220	129	21	344	343
16	159	47	134	9	175	138	54	283	14	206	324	305
17	273	98	75	118	335	69	173	228	1	220	314	159
18	215	11	17	249	225	29	212	204	30	325	303	204
19	185	38	13	253	203	131	306	113	58	275	267	308
20	150	74	45	136	62	103	174	100	108	256	283	346
21	70	245	201	192	40	261	169	237	66	142	337	328
22	35	101	226	82	201	181	39	168	123	85	354	336
23	155	102	99	12	191	194	54	267	3	187	345	349
24	232	94	206	53	296	49	190	279	4	231	338	358
25	85	23	27	361	167	31	156	257	42	315	311	365
26	362	20	15	341	234	139	288	164	80	310	306	364
27	285	44	87	134	96	182	148	177	241	278	319	360
28	240	198	179	105	51	235	133	236	117	224	340	359
29	57	366	287	7	248	184	66	247	114	145	356	282
30	107		63	22	251	148	110	318	19	229	352	290
31	299		88		297		178	293		309		353

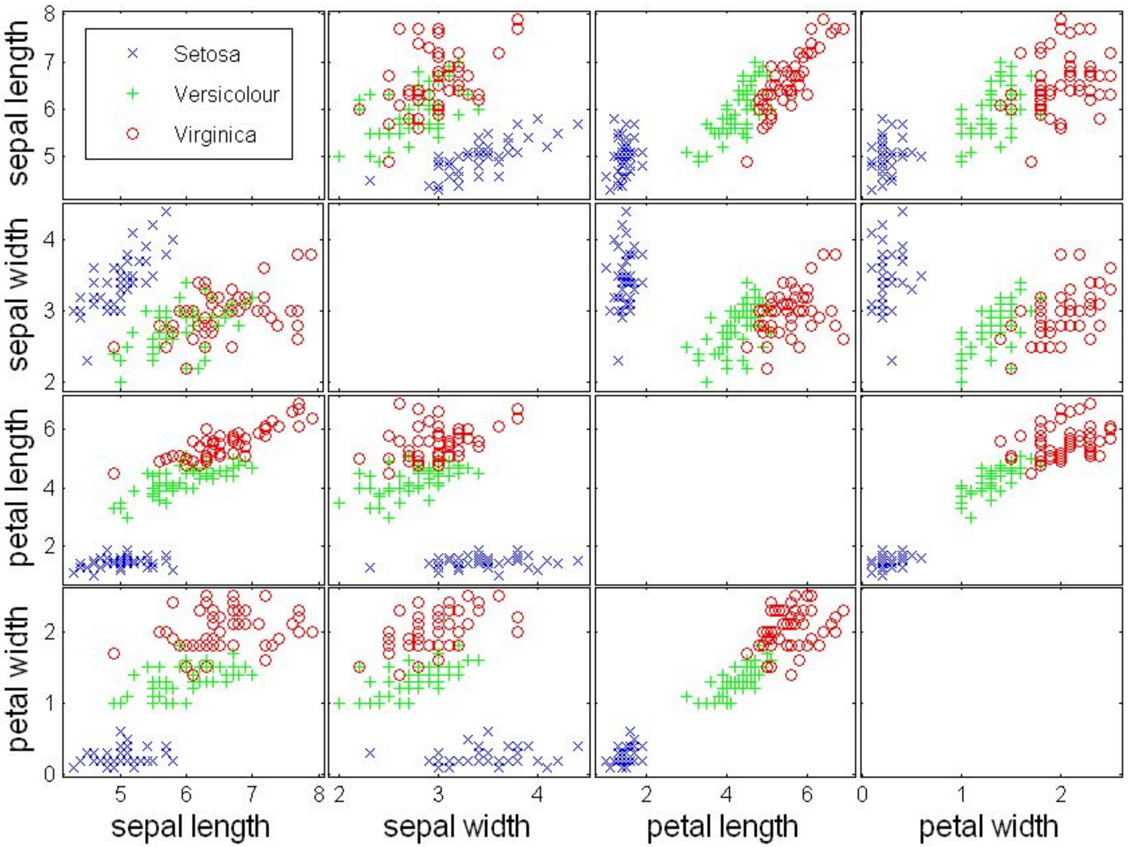
Source: ABS 2017

https://www.abc.net.au/news/2017-12-13/australias-most-and-least-popular-birthdays-revealed/9241978

Scatterplot matrix of iris data

- Plot each dimension against each other one

- Which one(s) give(s) the best data class separation?
- Look how nicely petal width (right) separates out the three colours, regardless of *y* axis



Scatterplot matrix considerations

- Create a matrix of scatter plots of all pairs of dimensions (variables)
- Can inspect **many relationships simultaneously**
- Convenient for **spotting correlation between some variables**
- Also for spotting outliers

Week 4 Lecture & Workshop 5 Alignment

- Explore preprocessing techniques:
  - **Tokenisation, casefolding, noise removal, stopword removal**
  - **Stemming vs lemmatisation**
  - How preprocessing choices affect downstream results
- Introduce text representations:
  - **Bag-of-Words**
  - **Sparse matrices**
  - Practice with **sklearn** and **nltk**
- Strengthen tabular data analysis and visualisation skills:
  - Use **pandas, matplotlib, and seaborn**
  - Create and interpret a range of plots
  - Improve readability and handle **overplotting**